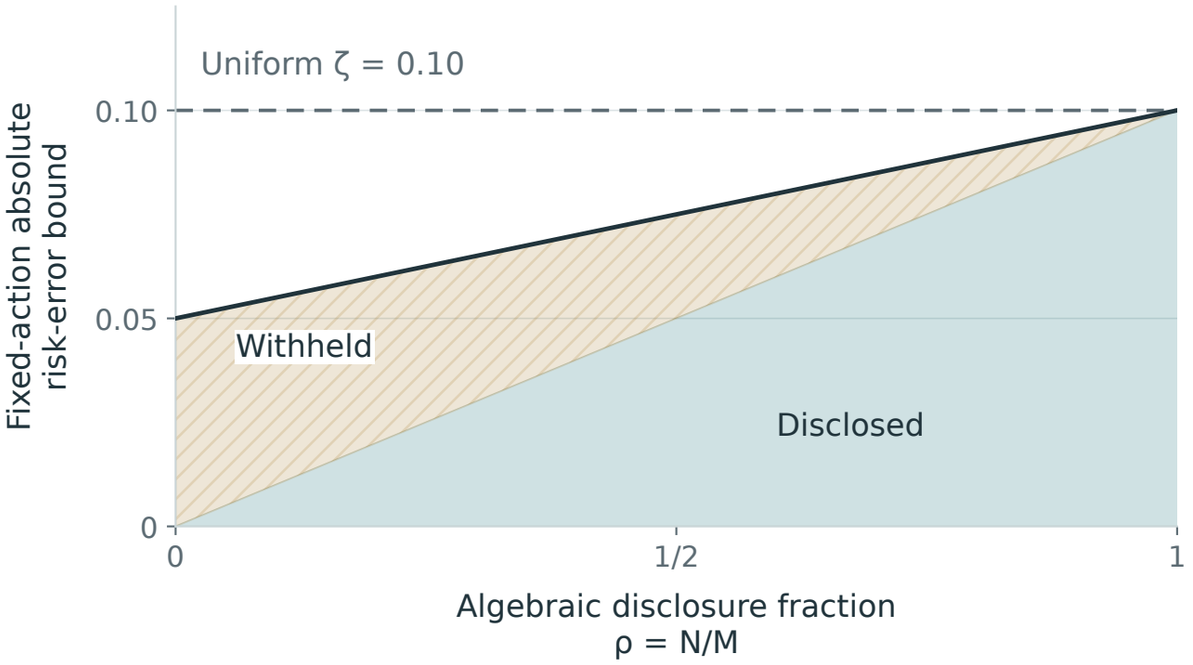
Two coefficient uncertainties contribute to risk

Illustrative simultaneous radii: τ = 0.25; Δb = 0.10; Δd = 0.20; η = 0.

A. Fixed-action envelope



Component upper bounds

Disclosed: $\rho\Delta_b$

Withheld: $(1-\rho)\tau\Delta_d$

Total: $0.05 + 0.05\rho$

B(0) = 0.05
Withheld distinctness

B(1) = 0.10
Disclosed direction

B. A separate regret argument

Same nonempty finite family F under exact and estimated objectives.
g*: exact minimizer; ĝ: estimated optimizer (η = 0).

Exact → estimated at ĝ	+	**Estimated optimization**	+	**Estimated → exact at g***
$R(\hat{g}) - \hat{R}(\hat{g}) \le \zeta = 0.10$		$\hat{R}(\hat{g}) - \hat{R}(g^*) \le \eta = 0$		$\hat{R}(g^*) - R(g^*) \le \zeta = 0.10$

Exact-risk regret $\le \zeta + \eta + \zeta = 0.20$

Algebraic envelope; no attained report path. Feasible N/M depends on n and the family.
Both stages use normalized risk units; regret requires the same targets, law, weights and nonempty finite family.
These are conditional bounds, not observed error or sampling confidence.